THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 2: **Power Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. 2^7 added 2 times equals: 1

(a) 2^8 **(b)** 4^7 **(c)** 2^{14} **(d)** 2^9

2. If $2^{2x+1} = 2^9$, the value of x is: 1

(a) 4 **(b)** 5 **(c)** 8 **(d)** 3

3. The number of digits in $2^4 \times 5^7$ is: 1

(a) 7 **(b)** 6 **(c)** 8 **(d)** 9

4. The ones digit of 7^{23} is: 1

(a) 3 **(b)** 9 **(c)** 5 **(d)** 7

5. If $3^x = 243$, then 3^{x-2} equals: 1

(a) 27 **(b)** 2187 **(c)** 241 **(d)** 9

6. The value of $(2^{-1} + 3^{-1})^{-1}$ is: 1

(a) $\frac{5}{6}$ **(b)** $\frac{1}{5}$ **(c)** $\frac{6}{5}$ **(d)** 6

7. Which of the following numbers is the greatest? 1

(a) 480×10^3 **(b)** 45×10^4 **(c)** 3.2×10^5 **(d)** 0.5×10^6

8. If $2^x = 5$, then 2^{x+3} equals: 1

(a) 256 **(b)** 13 **(c)** 15 **(d)** 40

9. 1

Assertion (A): $2^{-3} = \frac{1}{8}$.

Reason (R): For a non-zero number a, $a^{-n} = \frac{1}{a^n}$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** $(-1)^{101} = -1$. 1

Reason (R): Any non-zero number raised to the power 0 equals 1.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find x if $3^{2x-1} = 243$. 2

12. Write $(3 \times 10^5) + (4 \times 10^4)$ in scientific notation. 2

13. Find the ones digit of 2^{47}. 2

14. Find x if $2^{3x+2} = 16384$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A sheet of paper 0.1 mm thick is folded in half 10 times (assume this is possible). Find the thickness of the folded paper in mm and in cm. 3

16. Light travels 3×10^8 m in one second. How far does it travel in 2 minutes? Give the answer in scientific notation. 3

17. Find the value of $\dfrac{2^6 + 2^5}{2^4 \times (2+1)}$. 3

OR

A sheet of paper 0.2 mm thick is folded in half 8 times (assume this is possible). Find the thickness of the folded paper in mm and in cm. 3

18. Light travels 3×10^8 m in one second. How far does it travel in 5 minutes? Give the answer in scientific notation. 3

OR

Find the value of $\dfrac{3^5 + 3^4}{3^3 \times (3+1)}$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** The distance of the Earth from the Sun is about $150\,000\,000$ km and light travels at about 3×10^5 km per second. Students of a science club use these to answer questions on standard form. 5

(i) Write $150\,000\,000$ in standard form.

(ii) Find the time light takes to reach the Earth from the Sun.

(iii) Write that time in standard form.

(iv) Light takes about 1.3 s to reach the Earth from the Moon. Find the Moon's distance in standard form.

20. A culture has 2 bacteria and the number becomes 4 times every hour. Write the number after 3 hours as an expression with a power, find its value, and say how many times the original number it is. 5

OR

Body A has mass 9×10^{12} kg and body B has mass 3×10^9 kg. Which is heavier? How many times as heavy is it? Find the total mass in scientific notation. 5

END OF QUESTION PAPER — Total: 40 Marks